Do Wind-Directional Fire Features Improve Haze-Visibility Forecasts?

A Multi-Season Evaluation in Borneo and Sumatra

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Abstract

Satellite fire detections, weighted by wind direction, are commonly used in operational smoke prediction but are rarely evaluated on their own merit. Using public METAR archives, we created a dataset of twelve airports on Sumatra and Borneo where visibility was limited to less than 3 km due to hazy or smoky conditions (henceforth haze-limited visibility events). Then we investigated whether including wind-directional features of satellite fire detections (from the VIIRS active fire product) could improve 6 and 24 hour forecasts of haze-limited visibility events. To that end we built an eleven-year dataset (2015–2025) and used leave-one-year-out cross-validation to assess model skill. As controls, we used: time-shifted wind directions; randomised (permuted) wind directions; categories of wind reporting; and a model without wind-directional features (matched for missing data). Uncertainty was estimated with a block bootstrap at station and regional level. In single model runs, the directional model appeared to improve 24-hour forecasts substantially (+0.057 average precision), but this was driven by a lot of seed-to-seed variation in one extreme season, where different runs of the same model differ by almost 0.1 average precision. When we did five-seed ensembles, the directional model was only a few thousandths of average precision better than the permuted-wind placebo (+0.004) and the controls, with no consistently larger effects with vegetation fires only. The only noticeable benefit was that the fire count predictors helped 24-hour forecasts a little bit. Overall, we recommend seed ensembles and mask-preserving placebos for studies that claim directional wind-aware fire features are helpful.

1 Introduction

Peatland and forest fires in Indonesia produce some of the densest smoke on Earth. During strong El Niño dry seasons, haze from Sumatra and Kalimantan covers large parts of Equatorial Southeast Asia for weeks, closing schools and airports and causing severe respiratory illness [Huang et al., 2024]. Communities living around forests and peatland often learn that smoke has arrived only when it is already outside their window. Short-term haze forecasts that are local, open and honest about their uncertainty could help these communities prepare.

Satellite fire detections are the obvious predictor. NASA's Fire Information for Resource Management System (FIRMS) distributes VIIRS active-fire detections within hours of each overpass [NASA, 2026]. Smoke, however, travels with the wind. A detection 80 km upwind of a town matters more than one 80 km downwind. Wind-weighted fire-exposure features are therefore an intuitive and increasingly common design choice [Liao et al., 2023, Das et al., 2026], yet their *directional* component is rarely tested on its own. When a model with wind-weighted fire exposure beats a model without fire information, the gain may come from the fire amount,

*Code, data pipeline, protocol and review history: https://github.com/myudak/pavana/tree/main/research/directional-fire

distance or recency rather than from where the fires are relative to the wind. Other studies find that fire predictors add little once recent pollution and weather are known [Ganesh, 2026].

A second obstacle is the target. Hourly $PM_{2.5}$ observations over Kalimantan and most of Sumatra are short, sparse or unavailable before 2021. We instead use a long and openly archived proxy: *airport visibility*. Routine aviation weather reports (METAR) record horizontal visibility and present-weather codes for haze (HZ) and smoke (FU) every 30–60 minutes and have been archived for decades [Iowa Environmental Mesonet, 2026]. Visibility below 3 km with a haze or smoke code and no precipitation or fog is a direct, observed marker of smoke-haze impact.

We ask one question with a matched design. **Do wind-directional satellite fire features improve 6 h and 24 h haze-visibility forecasts beyond a non-directional fire-exposure feature with identical distance, age and fire-radiative-power weighting?** We also ask whether any gain survives realistic fire-data latency. Our contributions:

1. An eleven-year (2015–2025), twelve-airport benchmark of haze-limited visibility in Borneo and Sumatra, built entirely from keyless public archives, covering three major haze seasons (2015, 2019, 2023).

2. A matched ablation that isolates the directional term (directional vs non-directional exposure) against a strong station-state baseline that already includes the local wind direction, with time-shifted and mask-preserving permuted-wind placebos, missingness and wind-report-category controls. We show why rotated-wind placebos are not clean nulls.

3. Leave-one-year-out evaluation with station- and region-level block bootstrap, breakdowns by season and reporting convention, a vegetation-only sensitivity and a fire-data latency analysis, and evidence that single fits of tree models cannot resolve effects of this size in this setting.

4. An open implementation used in a public smoke-awareness application, Pavana, whose deployed model is the feature set evaluated here.

The answer is negative, and reaching it taught us two lessons about method. In a pre-specified comparison of single gradient-boosted fits, adding the directional exposure raised 24 h average precision from 0.417 to 0.474, a large and significant gain. Placebo winds reproduced almost all of it, and so did refitting the same model with another random seed: the differences between single fits were dominated by seed-to-seed variation in one extreme season. With five-seed ensembles, the directional model is at most a few thousandths of average precision above time-shifted and permuted-wind placebos and non-directional controls with the same missingness, with no consistent sign across aggregations, horizons or the two model families. Fire counts improve 24 h forecasts modestly. We think the checks we needed, seed ensembles and mask-preserving placebos, apply to any study that credits wind-weighted fire features.

The protocol, including the primary hypothesis, was written down and committed before any result reported here was computed (Appendix A). Changes made after seeing results are listed there and labelled in the text.

2 Related work

Haze forecasting in Equatorial Asia. Regional haze warnings rely on dispersion models and on satellite hotspot and smoke-plume analyses from the ASEAN Specialised Meteorological Centre. Process studies link the transport of Indonesian smoke to Singapore to south-easterly low-level jets and boundary-layer mixing [Huang et al., 2024]. On the data-driven side, Wang [2020] trained a convolutional network on 35 years of meteorological and hydrological fields to forecast severe haze at Singapore. They found skill above climatology but many missed events, and they used no fire data. Fire-to-PM modelling in Riau and Pontianak with gradient boosting

and ensembles exists [Faryuni and Sampurno, 2026], but it targets $PM_{2.5}$ at a single city and does not isolate directional fire information.

Visibility as an observed target. Airport visibility has long been used as a proxy for particulate pollution where monitors are scarce, including reconstructions of $PM_{2.5}$ from near-surface visibility across the Northern Hemisphere since 1959 [Hao et al., 2024]. In Central Kalimantan, visibility at Palangka Raya falls non-linearly with $PM_{2.5}$ and PM_{10}, dropping below 0.2 miles during the 2015 fires [Santoso et al., 2022]. Tree ensembles also forecast airport fog and low visibility from METAR and reanalysis [Cerda Castillo, 2025]. We use visibility with haze codes as the forecast target itself.

Wind-aware fire exposure. Weighting fire radiative power by distance and by the angle between wind and fire-to-receptor direction is an established feature-engineering choice. Liao et al. [2023] use inverse-distance and wind-angle weighting in graph-network $PM_{2.5}$ forecasts, and Das et al. [2026] evaluate distance- and direction-aware exposure metrics. Epidemiological work defines upwind exposure from wind direction and fire location [Aguilera et al., 2021]. In contrast, a recent site-level forecasting study found that inverse-distance fire features added nothing beyond prior-day $PM_{2.5}$ and weather [Ganesh, 2026]. To our knowledge, no study has isolated the *directional* term against a non-directional control with identical weighting, checked it with rotated-wind placebos, and repeated the test across several haze seasons and fire-data latencies. That is the gap this paper fills.

3 Data

Airport observations. We use METAR reports for twelve airports from the Iowa Environmental Mesonet archive [Iowa Environmental Mesonet, 2026]: four in Indonesian Kalimantan (WIOO Pontianak, WAOO Banjarmasin, WALL Balikpapan, WAQQ Tarakan), five in Malaysian Borneo (WBGG Kuching, WBGS Sibu, WBGR Miri, WBGB Bintulu, WBKK Kota Kinabalu), and three in Sumatra and the Riau Islands (WIBB Pekanbaru, WIPP Palembang, WIDD Batam) (Figure 1). Palangka Raya, in the most fire-affected province, has no archived METAR and is absent. We keep reports from 1 January 2015 to 31 December 2025, both routine and special reports (IEM report types 3 and 4). When an hour has several reports, the last one in that UTC hour is used, so a correction (COR, 0.8% of selected reports) replaces the report it corrects when both fall in the same hour; 62.0% of issue hours had more than one report. IEM distributes the reports with very little quality control [Iowa Environmental Mesonet, 2026], and we add only two rules: visibility is clipped at the 10 km reporting cap (which removes one erroneous 8000 km value), and a calm report's direction of 0° is treated as undefined rather than as a north wind.

Target. For each station-hour t with a report, the target at lead time $h \in \{6, 24\}$ h is a *haze-limited visibility event*: at $t + h$ the reported visibility is below 3 km, the present weather contains HZ or FU, and it contains no precipitation or fog code. Hours without a report at $t + h$ have an unknown target and are dropped, as are reports with visibility below 3 km whose present-weather group cannot be parsed. Test labels are never imputed.

The target is an observer-reported marker of haze, not a smoke measurement: HZ is also used for dust or urban pollution, and reporting practice differs by station and changes over time. The seasonality points mostly to fire smoke: 88.2% of event hours fall in the July–November burning season. Proximity to detections is a weak check here, because almost every hour has a VIIRS detection within 300 km in the preceding 72 h (99.7% of event hours, 93.2% of other hours). We therefore interpret all results as results about this visibility event, not about smoke concentration.

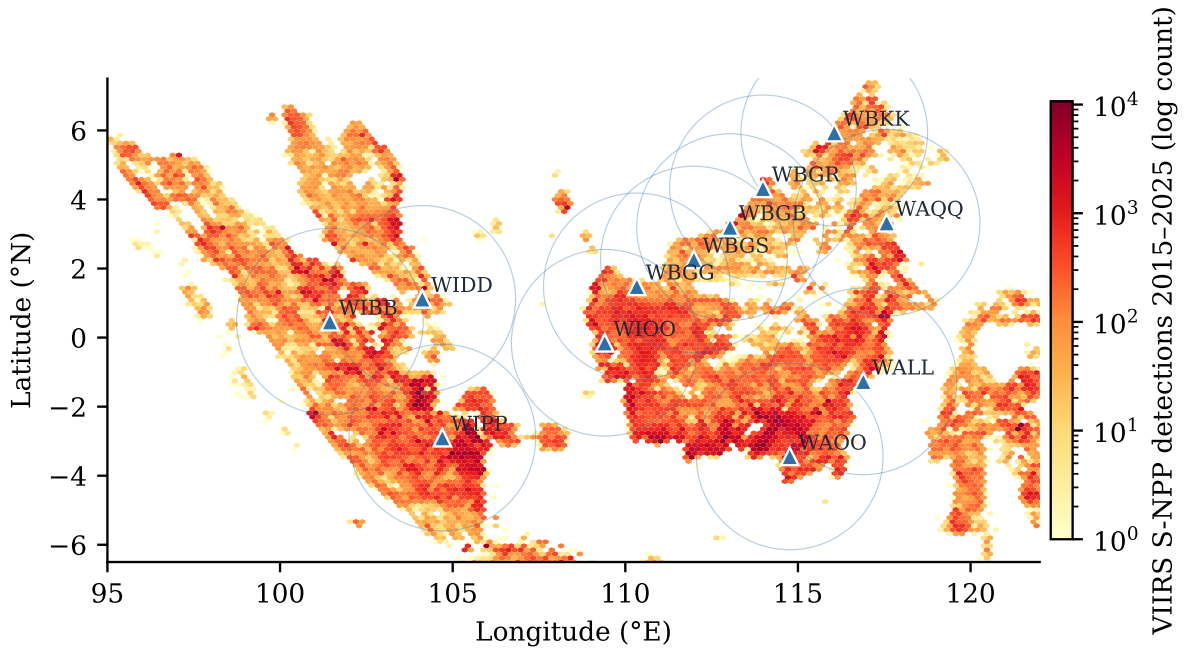


Figure 1: The twelve airports (triangles) with their 300 km fire-search radius, over the density of VIIRS S-NPP detections 2015–2025.

Coverage and reporting conventions. Report completeness differs sharply between countries (Figure 2). The Malaysian stations report almost every hour throughout. The Indonesian stations have long gaps before 2021, for example 12–15% of hours at Pontianak in 2016 and 2020. Coding habits also differ: Malaysian stations use HZ almost exclusively (FU in $< 0.1\%$ of reports), while Indonesian stations use FU more often. Our target accepts either code, and we report results separately by country.

Fire detections. We use VIIRS Suomi NPP active-fire detections from the FIRMS yearly country archives for Indonesia and Malaysia [NASA, 2026], restricted to 93–122°E, 9°S–9°N. A single sensor keeps the detection frequency consistent over the period; mixing in NOAA-20 (from 2018) and NOAA-21 (from 2023) would change the constellation over time. Every record carries the archive's `type` attribute, which the near-real-time feed lacks, so these are processed archive detections. Their locations and counts can differ from what was available in real time; the latency analysis below shifts detection times but cannot reproduce that difference. Detections are thermal anomalies, not confirmed fires, and their absence does not imply the absence of fire under cloud or thick smoke.

The main analysis uses all detection types, as the deployed model does. Non-vegetation sources (FIRMS types 1–3: volcanoes, static industrial sources such as gas flares, and offshore platforms) are a small share of the distance-weighted exposure at most airports but not at all: 54% at Bintulu, next to a liquefied-natural-gas plant, and 7% at Batam and 7% at Miri. A sensitivity analysis repeats the key comparisons with vegetation fires (type 0) only.

Haze seasons. Events are highly concentrated in time. Three seasons, 2015, 2019 and 2023, contain most event hours, with 2015 dominant at almost every station (Figure 2). Any single temporal split therefore scores very few independent episodes, which motivates our leave-one-year-out design (Section 4).

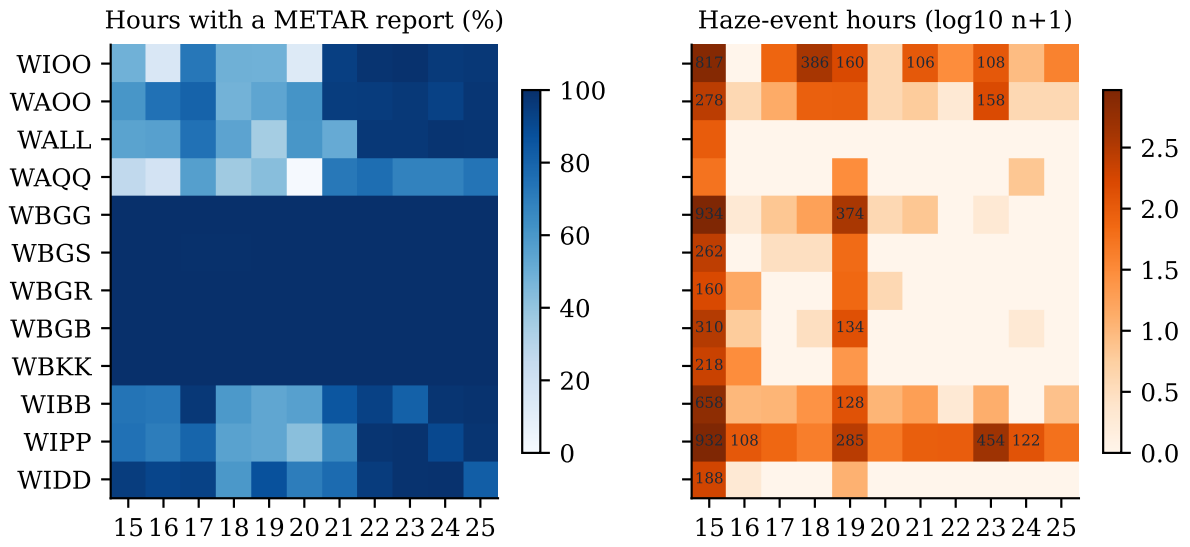


Figure 2: Left: share of hours with a METAR report per station and year. Right: haze-event hours per station and year (numbers shown where ≥ 100).

4 Methods

4.1 Features

All features at issue time t use only information available at t: station reports at or before t, and fire detections acquired at or before $t - L$, where L is the fire-data latency.

Station state (C). Current visibility, minimum visibility over the last 6 h and 24 h, current haze code, share of reports with a haze code over 24 h and 72 h, share with precipitation over 24 h, relative humidity, temperature, wind speed, **wind direction** (sine and cosine), local hour, day of year, and the station-month climatological event rate (estimated from training years only). Including the wind direction in the baseline means that a station's typical "haze wind" is already available to every model. A directional fire feature can only help by locating the fires relative to that wind.

Fire counts (D). Detections within 100 km in the last 24 h, and within 300 km in the last 72 h.

Fire exposure. For detections i within 300 km acquired in the 72 h before $t - L$, let d_i be the great-circle distance to the station, a_i the detection age, FRP_i its fire radiative power (1 MW when missing), $\hat{w}$ the unit vector of the wind's flow direction (the reported meteorological "from" direction plus 180°), and $\hat{h}_i$ the unit vector from the detection towards the station. We compare a non-directional exposure

$$S^{\mathrm{iso}} = \sum_i \mathrm{FRP}_i \, e^{-d_i/\lambda} \, e^{-a_i/\tau} \tag{1}$$

with a directional (upwind) exposure

$$S^{\mathrm{dir}} = \sum_i \mathrm{FRP}_i \, e^{-d_i/\lambda} \, \max(0, \hat{w} \cdot \hat{h}_i) \, e^{-a_i/\tau}, \tag{2}$$

with $\lambda = 50$ km and $\tau = 24$ h as in the deployed Pavana model (not tuned here). The only difference between the two is the alignment term.

Placebos. A placebo should keep everything about S^{dir} except its link to the actual wind–fire geometry. Rotating the wind by 180° or 90° is a natural candidate, but it is not a clean negative control. A rotated exposure is still computed from the current wind and the actual fire positions, so it carries geometric information correlated with S^{dir} (for example $S^{\text{dir}} + S^{180} = \sum_i w_i \, |\hat{w} \cdot \hat{h}_i|$, a measure of how aligned the fires are with the wind axis). It also shares the missingness pattern of S^{dir} exactly. We therefore report rotations only for contrast, and use two families of placebos that break the current wind–fire geometry:

- *Time-shifted wind* (S_k^{shift}): the same station's wind from k hours earlier, with $k \in \{72, 168, 336, 720\}$; $k = 168$ h is the pre-specified main placebo (same hour of day). A shifted wind can still share the regime of the current wind (monsoon flow persists for weeks) and has its own missingness pattern, so we report how similar the shifted and current winds are (Appendix B).
- *Mask-preserving placebos*, which have exactly the missingness of S^{dir}: directions permuted at random among the hours with a defined wind of the same station and calendar month (S^{perm}, three seeds), and the 168 h-shifted direction (the last defined value at or before $t - 168$ h) applied only where the current direction is defined ($S_{168}^{\text{shift,m}}$).

Wind-report categories. S^{dir} is undefined whenever the selected report has no usable direction. This happens for three different reasons with different meanings: calm wind (`00000KT`, 13.6% of issue hours), variable wind (`VRB`, 15.5%), and a missing wind group (0.1%). To separate missingness from direction we use two non-directional controls. $S_{\text{calm}}^{\text{iso}}$ is the non-directional exposure set to missing whenever the direction is undefined (the exact missingness of S^{dir}). The category control ("cat") gives the model indicators for the three categories and the non-directional exposure split by category ($S^{\text{iso}} \cdot \mathbb{1}[\text{category}]$), so any interaction between exposure and reporting state is available explicitly.

Persistent flow (exploratory). Smoke arriving over 72 h follows the persistent flow rather than one hourly surface report. We therefore also compute S^{dir} with the 24 h speed-weighted vector-mean wind (defined in 95% of hours), with its own shifted placebo.

4.2 Variants

Every variant is a LightGBM binary classifier [Ke et al., 2017] with the deployed hyperparameters (learning rate 0.05, 31 leaves, at least 200 rows per leaf, feature fraction 0.9, bagging fraction 0.8, $\ell_2 = 1$) and a fixed budget of 400 boosting rounds, so no variant receives more tuning than another. The random seed controls bagging and feature subsampling, and single fits turned out to differ by almost 0.1 AP between seeds in some test years (Section 5.2). Each LightGBM variant is therefore a five-seed ensemble: the mean predicted probability of five fits that differ only in the seed. The protocol specified a single fit; those results are reported separately. We also report three reference models: persistence (the event indicator at t), station-month climatology, and logistic regression on C.

4.3 Evaluation

Leave-one-year-out (primary). Each calendar year 2015–2025 is held out in turn. The model is trained on all other years, excluding 14 days on either side of the test year to limit leakage across boundaries. Pooled out-of-fold predictions are scored once per variant and lead time. Training on later years to predict an earlier one is not an operational forecast. It is a way to score several independent haze seasons, which is what the attribution question needs.

Variant	Features
C	station state
D	C + fire counts
D0	D + S^{iso}
E	D0 + S^{dir}
Rshift	D0 + S^{shift}_{168} (pre-specified main placebo)
Rshift$k$	D0 + S^{shift}_{k}, $k \in \{72, 336, 720\}$ h
Rperm1–3	D0 + S^{perm} (mask-preserving, three seeds)
Rshiftm168	D0 + $S^{\text{shift,m}}_{168}$ (mask-preserving)
R180, R90	D0 + S^{dir} with rotated wind (not clean nulls; for contrast)
D0m	D0 + $S^{\text{iso}}_{\text{calm}}$ (missingness of S^{dir}, no direction)
D0cat, Ecat	D0 / E + wind-report categories and category-split exposure
D0v, Ev, Rshiftv, D0mv	as D0, E, Rshift, D0m with vegetation fires only (FIRMS type 0)
Ep, Rpshift	D0 + persistent-flow S^{dir} / its shifted placebo (exploratory)
E_{served}	D + S^{dir} (the deployed feature set)

Table 1: LightGBM feature sets. The directional question is E against the placebos and the non-directional controls (D0m, D0cat), not E against D0 alone. Logistic-regression versions of C, D0, E, Rshift, Ep, Rpshift, D0cat, Ecat and Rperm1 (prefix "lg", with $\log(1 + x)$ exposures and missingness indicators) repeat the key comparisons in a second model family. Variants that were not in the frozen protocol (all except C, D, D0, E, R180, R90 and E_{served}) are listed with their dates in Appendix A.

Operational split (secondary). Train on 2015–2020 and test on 2023–2025, which is the configuration of the deployed model.

Latency. The main setting is $L = 3$ h, the nominal latency of FIRMS near-real-time data. Sensitivity runs use $L \in \{0, 6, 12, 24\}$ h, for D0, E and Rshift at 24 h (five-seed ensembles).

Metrics. Events are rare (0.2–3.7% of hours per station), so the primary metric is average precision (AP, area under the precision–recall curve). We also report ROC-AUC, the Brier skill score against station-month climatology, and the share of event hours captured when alerting on the top 2% of hours.

Pooled and per-season scores. Pooled out-of-fold AP is dominated by the largest season (2015), whose fold is also the hardest extrapolation. We therefore also report the mean of within-year ΔAP over the years with at least 100 event hours (ten of eleven years), which weights each season equally.

Uncertainty. Differences in AP are assessed with a paired block bootstrap (1000 resamples) over station × ISO-week blocks. This keeps within-week autocorrelation and the pairing of variants on identical rows. We report 95% percentile intervals. A haze episode often affects several airports at once, so station-level blocks can understate uncertainty. As a sensitivity check we repeat the key contrasts with blocks that hold all stations together: ISO weeks, calendar months, and calendar years (500 resamples each). Eleven years do not make eleven independent episodes, and the year-block intervals, from only eleven clusters, are coarse.

Hypotheses. H1 (primary): $\Delta\text{AP}(\text{E} - \text{D0}) > 0$ at $h = 24$ h, $L = 3$ h, under leave-one-year-out. H2: the directional gain is larger at 24 h than at 6 h. H3: placebos do not reproduce the gain. H4: the directional gain shrinks as L grows. All results are reported whatever their sign. Appendix A lists the deviations from the frozen protocol, in order: the shifted-wind placebo and missingness control, the persistent-flow variant, the logistic family and per-season scores, the

	$h = 24$ h				$h = 6$ h			
Model	AP	AUC	BSS	hit@2%	AP	AUC	BSS	hit@2%
Persistence	0.310	0.775	0.109	56.1	0.259	0.758	−0.014	52.3
Climatology	0.030	0.798	0.000	5.6	0.027	0.789	0.000	5.5
GBM C	0.428	0.936	0.282	64.0	0.486	0.937	0.324	71.1
GBM D	0.447	0.950	0.296	66.7	0.478	0.940	0.316	68.1
GBM D0	0.434	0.937	0.290	63.0	0.477	0.939	0.317	68.2
GBM E	0.451	0.950	0.301	66.9	0.500	0.960	0.330	71.5
GBM Rshift	0.448	0.950	0.299	66.8	0.497	0.960	0.328	71.4
GBM Rperm1	0.446	0.949	0.298	66.2	0.498	0.959	0.329	71.5
GBM D0m	0.449	0.950	0.299	66.9	0.497	0.958	0.328	71.2
GBM D0cat	0.449	0.949	0.299	67.0	0.496	0.957	0.327	71.1
GBM Ecat	0.452	0.949	0.302	66.7	0.479	0.940	0.318	68.7
Logit C	0.563	0.965	0.399	74.5	0.581	0.967	0.413	78.4
Logit D0	0.559	0.969	0.395	75.3	0.561	0.970	0.391	79.2
Logit E	0.562	0.970	0.398	75.4	0.562	0.970	0.393	79.2

Table 2: Pooled leave-one-year-out scores ($L = 3$ h). GBM rows are five-seed LightGBM ensembles. AP: average precision; AUC: ROC-AUC; BSS: Brier skill score against station-month climatology; hit@2%: share of event hours (%) caught when alerting on the top 2% of hours.

controls added in response to review (wind-report categories, mask-preserving placebos, further shifts, vegetation-only fires, regional bootstrap blocks), and the five-seed ensembles. All were added after first results and are labelled as such.

5 Results

Unless stated otherwise, results are for leave-one-year-out evaluation at $L = 3$ h, and every LightGBM variant is a five-seed ensemble (Section 4). The pooled 24 h test set has 919,470 station-hours and 7,655 event hours.

5.1 Baselines and model families

Table 2 and Figure 3 summarise the models. Every learned model is far above climatology (AP 0.030) and persistence (0.310) at 24 h. Persistence is better at 24 h than at 6 h: haze visibility has a strong diurnal cycle, so the same hour on the next day is a better analogue than six hours later.

The largest differences in the table are between model families, not between feature sets. Under leave-one-year-out, logistic regression on station state alone reaches AP 0.563 at 24 h, against 0.428 for gradient boosting on the same features. Most of that gap comes from the 2015 fold, the most extreme season, which tree models trained on milder years cannot extrapolate to. On the operational split the order reverses (Section 5.8), so neither family is uniformly better.

5.2 Single fits cannot resolve the effect (pre-specified analysis)

The protocol specified one LightGBM fit per variant. Under that design, adding the directional exposure raised pooled 24 h AP by +0.057 [0.030, 0.085] over D0, which would have confirmed H1 decisively. The shifted-wind placebo and the missingness control gained almost as much (Rshift−D0 +0.054, D0m−D0 +0.054). On closer inspection, nearly all of these pooled differences came from the 2015 fold. There the variants fell into two groups, with AP near 0.52 or near 0.59, and group membership did not follow feature content: Ecat, which contains every

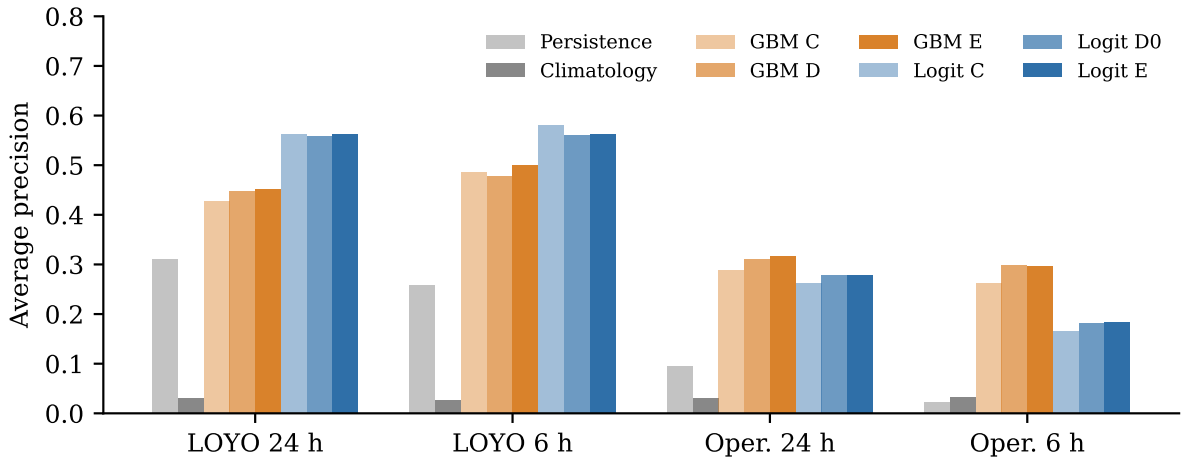


Figure 3: Average precision by model and evaluation setup ($L = 3$ h). Orange: gradient boosting (five-seed ensembles); blue: logistic regression; grey: persistence and climatology.

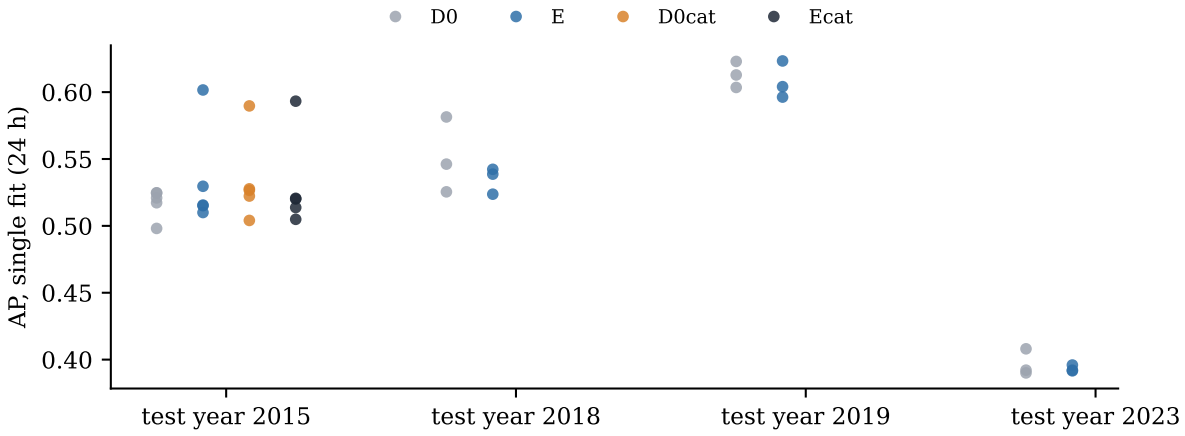


Figure 4: AP of single LightGBM fits that differ only in the random seed (24 h, $L = 3$ h, one test year each). Five seeds for 2015, three for the other years shown.

column of D0cat plus the directional exposure, landed in the low group while D0cat landed in the high one.

Refitting single folds with different seeds shows why (Figure 4). In the 2015 fold, D0 ranges from 0.498 to 0.525 and E from 0.510 to 0.602; the high outcome appears in about one fit in five, for any variant. Smaller seasons are also affected: in 2018, D0 alone ranges from 0.525 to 0.581. The seed controls bagging and feature subsampling, so a single fit is one draw from a wide distribution, and the differences we want to detect (a few thousandths of AP) are far smaller than its spread. The pre-specified single-fit comparison is therefore not informative about H1–H3, in either direction. We keep it in Appendix B and base the conclusions below on five-seed ensembles. Five seeds reduce but do not remove this noise (for example, Ecat at 6 h stays an outlier, Table 2, again because of the 2015 fold: 0.571 against 0.602 for D0cat), so small single contrasts should be read with that in mind.

5.3 The directional term with seed ensembles (H1, H3)

With seed ensembles, the directional exposure improves on the matched non-directional model by +0.017 [0.007, 0.027] at 24 h (Figure 5). Part of that is D0 being low rather than E being

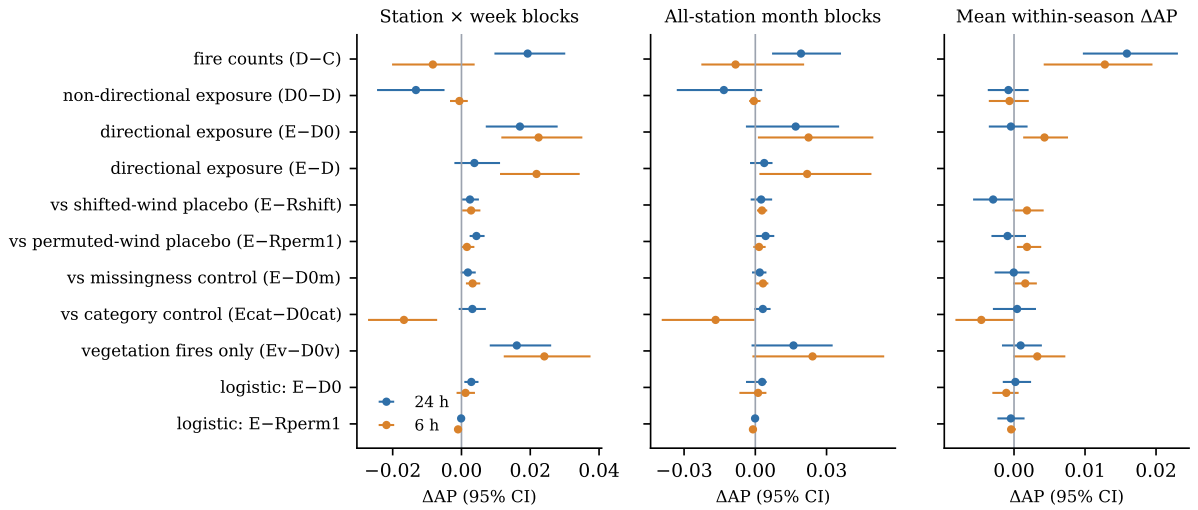


Figure 5: Differences in average precision under leave-one-year-out ($L = 3$ h; LightGBM rows are five-seed ensembles), with 95% block-bootstrap intervals (500 resamples). Left: station × ISO-week blocks. Middle: calendar-month blocks holding all stations together. Right: mean of within-year differences over the ten years with at least 100 event hours.

high: against the counts-only model, E gains +0.004 [−0.002, 0.011]. Against the controls, the margin shrinks to a few thousandths:

- placebos: E−Rshift +0.002 [0.000, 0.005] and E−Rperm1 +0.004 [0.002, 0.007];
- missingness controls: E−D0m +0.002 [0.000, 0.004], and with the wind-report categories and category-split exposure already in the model, adding the directional exposure changes AP by +0.003 [0.000, 0.007].

When each season is weighted equally the picture is the same: E−D0 0.000 [−0.004, 0.002], E−Rperm1 −0.001 [−0.003, 0.002]. Holding all stations together in calendar-month blocks, which respects the regional extent of haze episodes, widens the intervals enough that even E−D0 includes zero ([−0.004, 0.035]; E−Rperm1 [0.000, 0.008]). Year blocks, with only eleven clusters, are coarser still (Appendix B).

The logistic family, which has no seed, agrees: lgE−lgD0 +0.003 [0.001, 0.005] and lgE−lgRperm1 0.000 [−0.001, 0.001]. At 6 h the margins are of the same size but somewhat steadier: E is ahead of the missingness control and the shifted placebo by about 0.003 AP with intervals that exclude zero even with month blocks (E−D0m +0.003 [0.000, 0.005]; E−Rshift +0.003 [0.001, 0.005]), but not ahead of the permuted placebo (+0.002 [−0.001, 0.004]), and the logistic family shows nothing (lgE−lgRperm1 −0.001 [−0.002, 0.000]).

In short, the directional exposure ends up a few thousandths of AP above its controls, and not consistently. Some station-week intervals exclude zero (E−Rperm1 at 24 h), but the same contrast is negative or null when seasons are weighted equally, the placebo comparisons change sign between aggregations (E−Rshift per season −0.003 [−0.006, 0.000]), the logistic family shows nothing against the permuted placebo, and the differences are smaller than the seed noise that remains in five-seed ensembles (Ecat−D0cat at 6 h is −0.017). The upper ends of the intervals against the placebos and controls stay below 0.01 AP under every block design. We therefore read any directional contribution as at most about 0.01 AP, plausibly a few thousandths at 6 h, and not established at either horizon. H1, stated against D0 alone, is supported in the pooled score, but H3 fails: the placebos and non-directional controls reach nearly the same skill, so the gain over D0 cannot be attributed to direction.

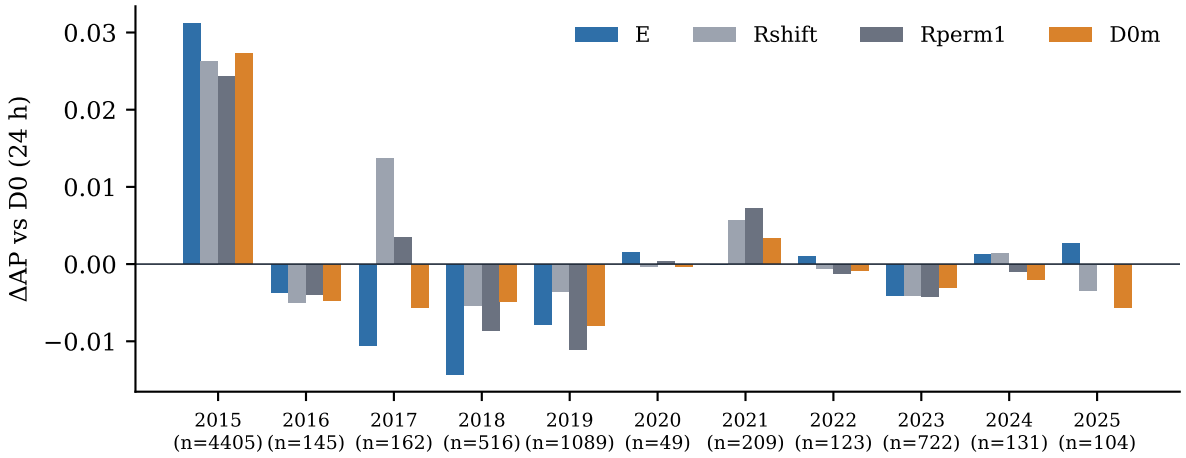


Figure 6: Within-year change in AP relative to D0 at 24 h for the directional exposure (E), the two placebos and the missingness control, all five-seed ensembles. n: event hours.

5.4 Fire counts

Detection counts improve on station state at 24 h: D−C +0.019 [0.010, 0.030] pooled and +0.016 [0.010, 0.023] per season. With month blocks the pooled interval is [0.007, 0.036]. At 6 h, where persistence carries most of the signal, the pooled effect is not distinguishable from zero (−0.008 [−0.020, 0.004]), although the per-season mean is positive (+0.013 [0.004, 0.019]). The non-directional exposure adds nothing on top of the counts when seasons are weighted equally (D0−D −0.001 [−0.004, 0.002]); pooled, it is slightly worse (−0.013 [−0.024, −0.005]), again because of the 2015 fold.

5.5 Where differences arise

Figure 6 splits the 24 h comparison by test year. The 2015 fold still carries the largest differences, now of similar size for E and for every control (+0.031 for E, +0.026 for Rshift, +0.027 for D0m), because D0 itself is low in that year (0.533, against 0.559 for D and 0.540 for C). In the other two major seasons E is slightly below D0 (2019: −0.008; 2023: −0.004). By reporting convention, E−D0 is +0.022 at the Indonesian stations (FU coding) and +0.013 at the Malaysian stations (HZ coding).

5.6 Vegetation fires only

Removing non-vegetation detections (gas flares, other static sources, offshore platforms) leaves the conclusions unchanged: D0v scores 0.434 against 0.434 for D0, and the directional term adds +0.016 [0.008, 0.025], of the same size as with all detections.

5.7 Lead time (H2) and latency (H4)

With no directional contribution detectable at either horizon, H2 (a larger directional gain at 24 h than at 6 h) has nothing to compare. For latency, the five-seed ensembles give E−D0 at 24 h of +0.017 with no latency, +0.017 at 3 h, +0.015 at 12 h and +0.016 at 24 h, and E−Rshift of +0.002, +0.002, +0.001 and +0.002 (Figure 7). Both curves are flat: the margin over the placebo stays within a few thousandths of AP whether the fire data are fresh or a day old, so there is no directional gain whose decay H4 could describe.

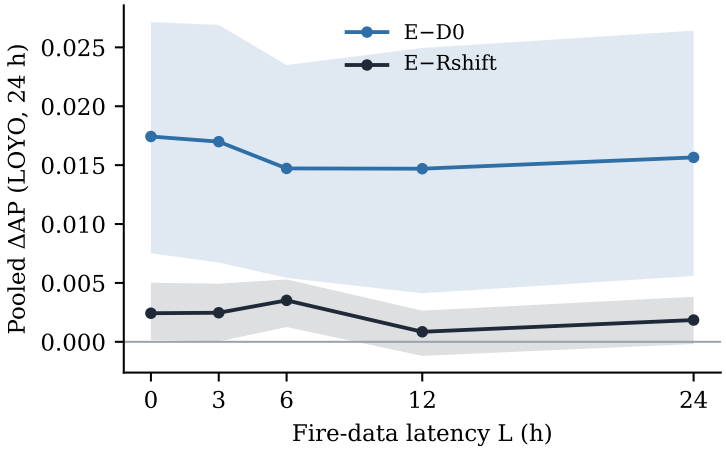


Figure 7: Pooled ΔAP at 24 h under leave-one-year-out as a function of fire-data latency L (five-seed ensembles), with 95% block-bootstrap bands (300 resamples).

5.8 Operational split

On the deployed configuration (train 2015–2020, test 2023–2025), gradient boosting is ahead of logistic regression (AP 0.311 for counts against 0.263 for logistic regression on station state at 24 h). Fire counts help (D−C +0.023 [0.007, 0.042]), and the directional term again matches its controls (E−Rperm1 0.000 [−0.005, 0.004]; E−D0m −0.002 [−0.005, 0.002]).

6 Discussion

No detectable directional contribution. A wind-weighted fire-exposure feature looks like a direct encoding of smoke transport. In our data, once seed noise is averaged out, it is at most a few thousandths of average precision better than the same exposure combined with a wind from another time, with directions permuted within the month, or with no direction at all but the same missingness, and not consistently so. Within the precision of an eleven-season record, the alignment between the reported wind and the fire positions adds little or nothing that these controls do not also provide. This is a statement about this feature, this target and this evaluation, not a proof that direction never matters for smoke.

Single fits of tree models are not enough. Our first, pre-specified analysis compared single LightGBM fits and found a large, significant directional gain, which several placebos then appeared to reproduce. We first read that as a mechanism: fire exposure behaving differently in calm, stagnant air. It was neither. The differences between variants were mostly differences between random seeds, concentrated in one extreme test year that the trees had to extrapolate to. A paired bootstrap does not catch this, because it resamples test rows for fixed fitted models and says nothing about how much the fitted models themselves would change with another seed. For rare, clustered events evaluated across seasons, we recommend at least a seed ensemble, or a report of the spread over seeds, before comparing feature sets.

Placebo design. The obvious placebo, rotating the wind, is not a clean null: it keeps the current wind axis and the exact missingness pattern of the directional feature. Time-shifted winds are better but not independent of the current wind. The 168 h-shifted direction still has a mean cosine similarity of 0.39 with the current one, and even 720 h earlier it is 0.36, because monsoon flow persists for weeks; its missingness matches the current report in only 69% of hours. Permuting directions within the month keeps the missingness exactly and lowers

the similarity to 0.19. We recommend mask-preserving placebos of this kind, together with a missingness-matched non-directional control, for any study that credits wind-aware fire features.

Why direction may not matter here. Several explanations are plausible, and our data cannot separate them. The airport's surface wind is a poor proxy for transport in the boundary layer, especially at night and near the coast, where sea breezes dominate several of our stations. Smoke from large peat fires is regional: in strong seasons the whole region is covered, so the direction of the nearest fires matters less than the total. The 300 km and 72 h window is long relative to how fast surface winds change. The persistent-flow variant was meant to address the last two points; it was run only as single fits and is not interpretable on its own. Trajectory- or reanalysis-based transport features are the natural next test.

Model families depend on the split. Logistic regression on station state beats every gradient-boosted model under leave-one-year-out, yet on the operational split gradient boosting is ahead (AP 0.311 for counts against 0.263). The ranking depends on whether the test period lies inside or outside the range seen in training. Leave-one-year-out trains on later years to score earlier ones, so it measures generalisation across seasons, not operational skill; for deployment decisions the forward-in-time split is the more relevant one.

Implications for Pavana. The deployed model uses detection counts plus the directional exposure. Our results support keeping the counts and suggest the directional term can be dropped with no measurable loss. We report this as found and have not changed the deployed system as part of this study.

Limitations. Visibility with a haze or smoke code is an observer-reported marker, not a smoke measurement; HZ is also used for dust and urban pollution, coding differs between countries and changes over time, and IEM applies little quality control. Several Indonesian stations have long reporting gaps before 2021, and Palangka Raya, in the most fire-affected province, is absent. We use one satellite sensor and processed archive detections rather than the near-real-time records that were available operationally; our latency shift moves detection times but cannot reproduce differences between the two versions. The exposure kernel is fixed from the deployed model; other distance or age scales might behave differently. Detections are not confirmed fires, and their absence under thick smoke or cloud biases exposure downward exactly when haze is worst. Bootstrap intervals treat the fitted models as fixed, and five seeds leave residual seed noise; intervals with all-station blocks are wider than the station-level ones, and eleven years are far from eleven independent episodes. Finally, most controls in this paper were added after the first results (Appendix A); they were added to test the first result, and they removed it.

7 Conclusion

Across eleven years and twelve airports in Borneo and Sumatra, satellite fire counts modestly improve 24 h forecasts of haze-limited visibility, but wind-directional fire exposure adds at most a few thousandths of average precision, inconsistently, over a time-shifted or permuted-wind placebo or a non-directional exposure with the same missingness. A single-fit comparison would have suggested a large directional gain; it was seed noise in one extreme season. Studies of wind-aware fire features should average over seeds and compare against mask-preserving placebos before attributing skill to direction.

Data and code availability

All inputs are public and need no keys: METAR reports from the Iowa Environmental Mesonet [Iowa Environmental Mesonet, 2026] and VIIRS S-NPP yearly active-fire archives from NASA FIRMS [NASA, 2026]. The data snapshot used here (built 16 October 2026) is fixed by file checksums. Feature construction, model training, the statistical analysis, every figure and every number quoted in the text are generated by scripts in the accompanying repository; numbers in the manuscript are inserted from the result tables, not typed. The exposure kernel and wind conversions are shared with the open-source Pavana application, whose TypeScript and Python implementations are tested against common vectors. Everything is available at https://github.com/myudak/pavana/tree/main/research/directional-fire, including the frozen protocol with its dated change log and the response to review.

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A Protocol and deviations

The study protocol was committed on 17 October 2026, before any result in this paper was computed. It fixed the question, data snapshot, target, nested feature groups (C, D, D0, E, R180, R90), fire-data latencies, model and hyperparameters (one LightGBM fit with fixed seed and 400 rounds for every variant), leave-one-year-out and operational evaluation, metrics, the block bootstrap and hypotheses H1–H4. Changes made afterwards, in order:

1. *hit@2% ties (17 Oct).* Binary predictors such as persistence produced ties that flagged more than 2% of hours. Exactly k hours are now alerted, ties broken at random. AP, ROC-AUC and Brier scores are unaffected.

2. *Persistent flow (17 Oct, exploratory).* Variants Ep and Rp180, using the 24 h vector-mean wind. Single fits only.

3. *Shifted-wind placebo (18 Oct, after the first leave-one-year-out results).* Rshift and Rpshift (wind from 168 h earlier), the logistic-regression family, and per-season mean ΔAP; the missingness control D0m.

4. *Review round 1 (25 Oct, before computing them).* Wind-report categories (D0cat, Ecat), mask-preserving placebos (Rperm1–3, Rshiftm168), further shifts (72, 336, 720 h) with similarity diagnostics, vegetation-only fires (D0v, Ev, Rshiftv, D0mv), all-station bootstrap blocks, and logistic versions lgD0cat, lgEcat, lgRperm1.

5. *Seed ensembles (25 Oct, after item 4's results).* Single fits were found to vary strongly with the seed (Section 5.2). Main LightGBM results now use five-seed ensembles for C, D, D0, E, Rshift, Rperm1, D0m, D0cat, Ecat, D0v and Ev, and for the latency runs. Single-fit results are kept below.

The decision rule for the directional claim did not change: direction counts as helping only if E beats the placebos and the non-directional controls, not only D0.

B Additional results

	single fit		five-seed ensemble	
Contrast (24 h)	pooled	per-season	pooled	per-season
D−C	+0.004	+0.013	+0.019	+0.016
D0−D	0.000	−0.001	−0.013	−0.001
E−D0	+0.057	+0.006	+0.017	0.000
Rshift−D0	+0.054	+0.009	+0.015	+0.003
E−Rshift	+0.003	−0.002	+0.002	−0.003
E−Rperm1	+0.009	−0.001	+0.004	−0.001
D0m−D0	+0.054	+0.006	+0.015	0.000
E−D0m	+0.003	0.000	+0.002	0.000
D0cat−D0	+0.049	+0.005	+0.015	0.000
Ecat−D0cat	−0.050	−0.009	+0.003	0.000
Ev−D0v	+0.053	+0.006	+0.016	+0.001

Table 3: The same contrasts as single LightGBM fits (the pre-specified design) and as five-seed ensembles, leave-one-year-out, 24 h, $L = 3$ h. Single fits show large pooled gains for E and for its controls alike; ensembles show none beyond the controls.

Contrast (24 h)	station × week	all-station week	all-station month	year
D−C	[0.010, 0.030]	[0.007, 0.032]	[0.007, 0.036]	[0.009, 0.028]
E−D0	[0.007, 0.028]	[0.005, 0.027]	[−0.004, 0.035]	[−0.007, 0.030]
E−D	[−0.002, 0.011]	[0.000, 0.007]	[−0.002, 0.007]	[−0.003, 0.006]
E−Rshift	[0.000, 0.005]	[−0.001, 0.005]	[−0.002, 0.007]	[−0.005, 0.005]
E−Rperm1	[0.002, 0.007]	[0.001, 0.007]	[0.000, 0.008]	[−0.002, 0.007]
E−D0m	[0.000, 0.004]	[−0.001, 0.004]	[−0.001, 0.005]	[−0.002, 0.003]
lgE−lgD0	[0.001, 0.005]	[0.000, 0.005]	[−0.004, 0.005]	[−0.004, 0.004]

Table 4: 95% intervals for pooled ΔAP under different bootstrap blocks (500 resamples; five-seed ensembles). The year column has only eleven clusters.

	72 h	168 h	336 h	720 h	permuted
Mean $\cos(\theta_t - \theta_{t-k})$	0.41	0.39	0.38	0.36	0.19
Missingness agrees (%)	70	69	68	67	100
AP, single fit (24 h)	0.468	0.471	0.465	0.467	0.465

Table 5: How similar the placebo winds are to the current wind (mean over stations), and the single-fit AP of each placebo variant. Other single-fit placebo variants: Rperm2 0.470, Rperm3 0.472, Rshiftm168 0.473; E 0.474, D0 0.417. Single fits are not comparable to each other at this precision (Section 5.2).